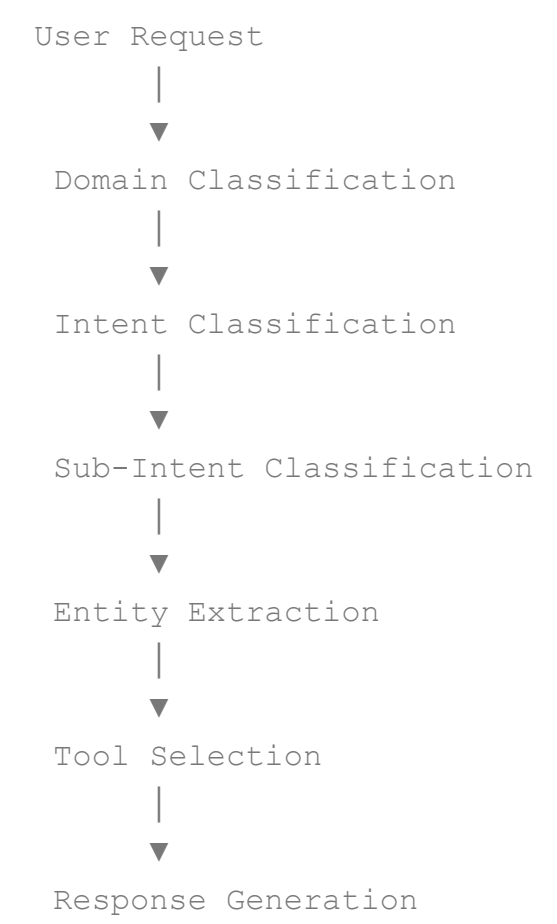


Noah Intent Taxonomy & Structured JSON Schema

Overview

Rather than treating every user request as an independent intent, Noah follows a **hierarchical intent classification** architecture. This improves scalability, simplifies supervised learning, and enables future expansion without retraining the entire intent space.

The prediction pipeline is:



Intent Hierarchy

1. WALLET

Responsible for all loyalty card and wallet operations.

Intent	Sub-Intent	Tool
WALLET	OPEN_CARD	wallet_tool
WALLET	SHOW_BARCODE	wallet_tool
WALLET	SHOW_POINTS	wallet_tool
WALLET	SHOW_REWARDS	wallet_tool
WALLET	ADD_CARD	wallet_tool
WALLET	REMOVE_CARD	wallet_tool
WALLET	LIST_CARDS	wallet_tool

2. PRODUCT

Responsible for discovering products.

Intent	Sub-Intent	Tool
PRODUCT	SEARCH	merchant_search
PRODUCT	SEARCH_BY_PRICE	merchant_search
PRODUCT	SEARCH_BY_BRAND	merchant_search
PRODUCT	SEARCH_BY_CATEGORY	merchant_search
PRODUCT	COMPARE_PRODUCTS	merchant_search
PRODUCT	CHECK_AVAILABILITY	merchant_search

3. MERCHANT

Responsible for merchant discovery.

Intent	Sub-Intent	Tool
MERCHANT	SEARCH	merchant_api
MERCHANT	NEARBY	merchant_api
MERCHANT	DETAILS	merchant_api
MERCHANT	OPENING_HOURS	merchant_api

Intent	Sub-Intent	Tool
MERCHANT	CONTACT	merchant_api

4. OFFER

Responsible for offers and discounts.

Intent	Sub-Intent	Tool
OFFER	SHOW_OFFERS	offers_tool
OFFER	SHOW_DISCOUNTS	offers_tool
OFFER	CASHBACK	offers_tool
OFFER	WEEKLY_FLYER	offers_tool

5. NAVIGATION

Responsible for routing.

Intent	Sub-Intent	Tool
NAVIGATION	ROUTE	maps_tool
NAVIGATION	DIRECTIONS	maps_tool
NAVIGATION	OPEN_MAP	maps_tool

6. RECOMMENDATION

Interfaces with the recommendation engine.

Intent	Sub-Intent	Tool
RECOMMENDATION	PRODUCT	recommendation_engine
RECOMMENDATION	MERCHANT	recommendation_engine
RECOMMENDATION	OFFER	recommendation_engine
RECOMMENDATION	PERSONALIZED	recommendation_engine

7. ACCOUNT

Responsible for user information.

Intent	Sub-Intent	Tool
ACCOUNT	PROFILE	firestore
ACCOUNT	PURCHASE_HISTORY	firestore
ACCOUNT	VISITS	firestore
ACCOUNT	SETTINGS	firestore

8. SHOPPING

High-level shopping assistant.

Intent	Sub-Intent	Tool
SHOPPING	BUILD_LIST	planner
SHOPPING	FIND_CHEAPEST_BASKET	planner
SHOPPING	OPTIMIZE_ROUTE	planner
SHOPPING	SPLIT_BASKET	planner

9. KNOWLEDGE

Questions answered using Noah's knowledge base.

Intent	Sub-Intent	Tool
KNOWLEDGE	FAQ	rag_engine
KNOWLEDGE	PAYTO_INFORMATION	rag_engine
KNOWLEDGE	APP_HELP	rag_engine

10. SYSTEM

Application-related requests.

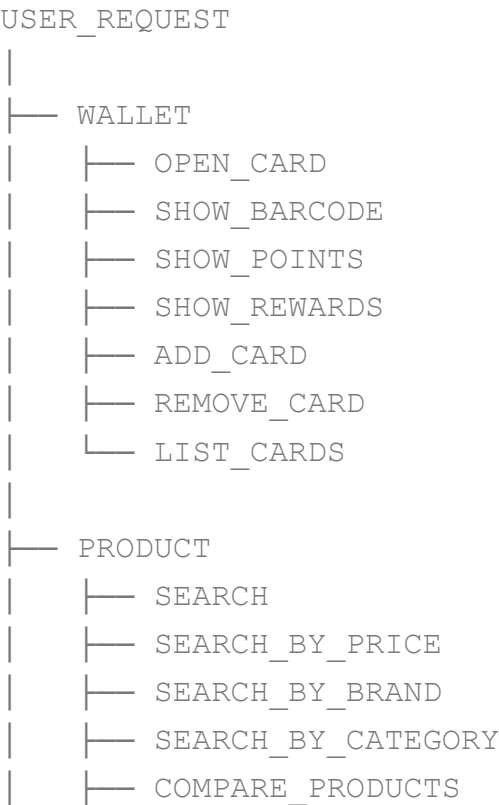
Intent	Sub-Intent	Tool
SYSTEM	HELP	planner
SYSTEM	REPORT_BUG	planner
SYSTEM	FEEDBACK	planner

11. CHAT

General conversational intents.

Intent	Sub-Intent	Tool
CHAT	GREETING	none
CHAT	GOODBYE	none
CHAT	THANKS	none
CHAT	SMALL_TALK	none
CHAT	UNKNOWN	none

Overall Architecture



└─ CHECK_AVAILABILITY

└─ MERCHANT

└─ SEARCH

└─ NEARBY

└─ DETAILS

└─ OPENING_HOURS

└─ CONTACT

└─ OFFER

└─ SHOW_OFFERS

└─ SHOW_DISCOUNTS

└─ CASHBACK

└─ WEEKLY_FLYER

└─ NAVIGATION

└─ ROUTE

└─ DIRECTIONS

└─ OPEN_MAP

└─ RECOMMENDATION

└─ PRODUCT

└─ MERCHANT

└─ OFFER

└─ PERSONALIZED

└─ ACCOUNT

└─ PROFILE

└─ PURCHASE_HISTORY

└─ VISITS

└─ SETTINGS

└─ SHOPPING

└─ BUILD_LIST

└─ FIND_CHEAPEST_BASKET

└─ OPTIMIZE_ROUTE

└─ SPLIT_BASKET

└─ KNOWLEDGE

└─ FAQ

└─ PAYTO_INFORMATION

└─ APP_HELP

└─ SYSTEM

- | — HELP
- | — REPORT_BUG
- | — FEEDBACK
- |
- └ CHAT
 - | — GREETING
 - | — GOODBYE
 - | — THANKS
 - | — SMALL_TALK
 - | — UNKNOWN

Final JSON Output Schema

Every request processed by Noah should produce a structured JSON object. This serves as the contract between the LLM, the Planner, and downstream tools.

```
{
  "domain": "PRODUCT",
  "intent": "SEARCH",
  "sub_intent": "SEARCH_BY_PRICE",

  "entities": {
    "product": "Frozen Pizza",
    "merchant": "REWE",
    "brand": null,
    "category": "Frozen Food",
    "price_min": null,
    "price_max": 2.00,
    "location": "Hamburg",
    "radius_m": 1000
  },

  "tool": "merchant_search",

  "response_mode": "hybrid",

  "requires_memory": false,
  "requires_rag": false,
  "requires_recommendation": false,

  "confidence": 0.98,
```

```
"reasoning": "User requests a product search with a maximum price const  
  
"planner_action": "SEARCH_PRODUCT_BY_PRICE"  
}
```

Response Modes

Response Mode	Description
<code>functional</code>	Opens an app screen or executes an action (e.g., display a barcode, open wallet).
<code>text</code>	Returns only natural language.
<code>hybrid</code>	Combines generated text with structured UI elements such as product lists or merchant cards.
<code>navigation</code>	Opens Google Maps or an in-app navigation interface.
<code>recommendation</code>	Displays results generated by the recommendation engine.

Example 1

User

Show me my Payback barcode.

```
{  
  "domain": "WALLET",  
  "intent": "OPEN_CARD",  
  "sub_intent": "SHOW_BARCODE",  
  "entities": {  
    "provider": "Payback"  
  },  
  "tool": "wallet_tool",  
  "response_mode": "functional",  
  "requires_memory": false,  
  "requires_rag": false,  
  "requires_recommendation": false,  
}
```



```
"confidence": 0.99,  
"planner_action": "DISPLAY_BARCODE"  
}
```

Example 2

User

Where can I buy frozen pizzas for less than €2 near me?

```
{  
  "domain": "PRODUCT",  
  "intent": "SEARCH",  
  "sub_intent": "SEARCH_BY_PRICE",  
  "entities": {  
    "product": "Frozen Pizza",  
    "price_max": 2.00,  
    "location": "CURRENT_LOCATION",  
    "radius_m": 1500  
  },  
  "tool": "merchant_search",  
  "response_mode": "hybrid",  
  "requires_memory": false,  
  "requires_rag": false,  
  "requires_recommendation": true,  
  "confidence": 0.97,  
  "planner_action": "SEARCH_PRODUCT_BY_PRICE"  
}
```

Example 3

User

Take me to the nearest Lidl.

```
{
  "domain": "NAVIGATION",
  "intent": "ROUTE",
  "sub_intent": "OPEN_MAP",
  "entities": {
    "merchant": "Lidl"
  },
  "tool": "maps_tool",
  "response_mode": "navigation",
  "requires_memory": false,
  "requires_rag": false,
  "requires_recommendation": false,
  "confidence": 0.99,
  "planner_action": "OPEN_GOOGLE_MAPS"
}
```

Design Principles

- **Domain** identifies the functional area of the application.
- **Intent** defines the user's primary objective.
- **Sub-Intent** refines the requested operation.
- **Entities** capture all relevant parameters extracted from the user's request.
- **Tool** specifies the backend component responsible for execution.
- **Response Mode** determines how the Flutter application presents the result.
- **Planner Action** is the normalized command that the planner executes.
- **Confidence** provides a measure of certainty and can be used to trigger clarification when below a predefined threshold (e.g., 0.70).